

ADVANCED MACHINE LEARNING · WORKSHEET 11

Dimensionality Reduction and PCA

Feature Extraction · Covariance · Eigenvectors · Principal Components

Lecture 11

Name:	Roll No.:	Date:
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PART I: WHY DIMENSIONALITY REDUCTION?

When a dataset has many features, not all of them are equally useful. Some carry important signal while others add noise and computational cost. How do we keep what matters?

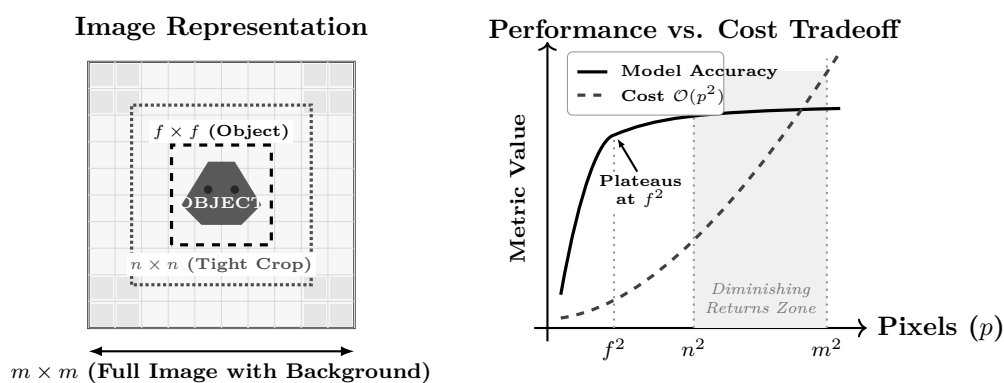
Many pixels, little additional signal

▷ HOOK

In Lecture 10, we saw that adding more features can increase variance and harm generalization. Feature selection answered: “Which original columns should remain?” But what if the information is spread across several columns and no single column captures it alone?

Suppose we are building an AI model that identifies the shape of an object present in an image. The image has size $m \times m$ pixels. Consider the following question:

- Should we feed the complete $m \times m$ image to the model?
- Should we instead use a tightly cropped $n \times n$ image?
- Or should we use a perfectly cropped $f \times f$ image that contains only the object?

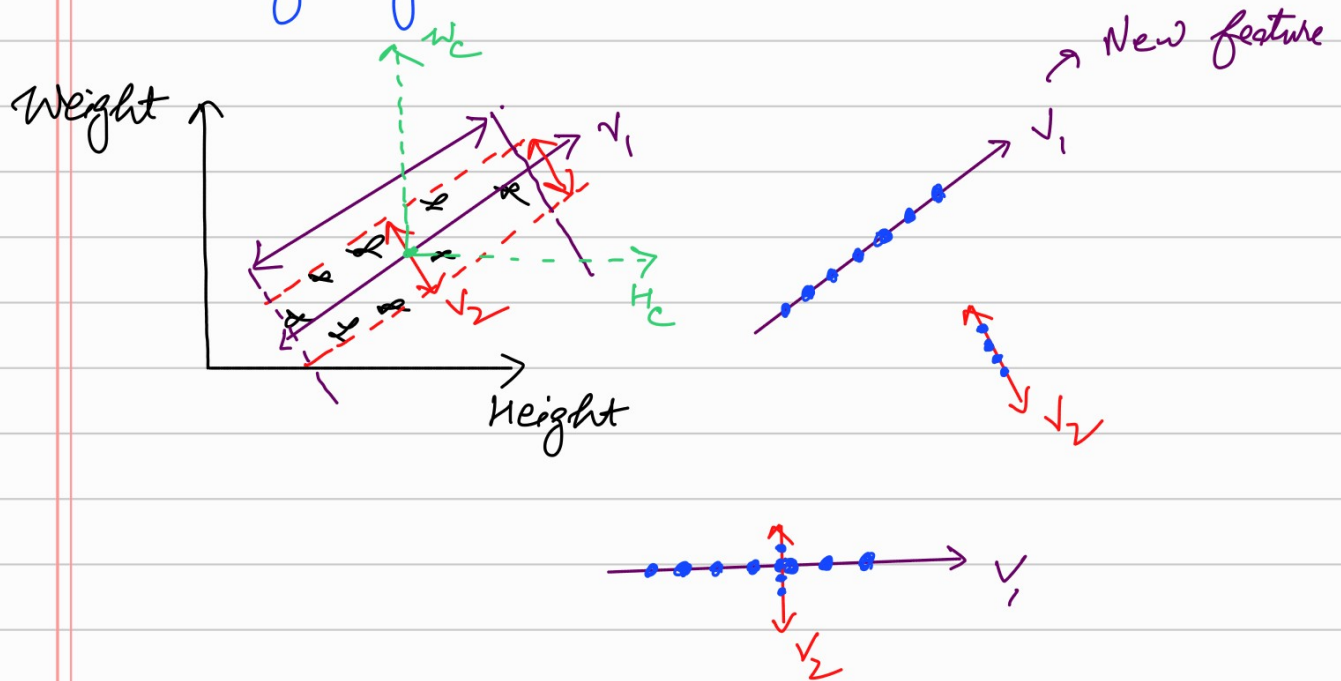


Left: An $m \times m$ image with a central object and crop regions. Right: Model performance saturates once the object is captured (f^2), while computational cost continues to explode as uninformative background pixels (m^2) are included.

Classroom question: If the full $m \times m$ image gives nearly the same accuracy as the $f \times f$ crop but requires far more computation, what are the additional pixels contributing?

PCA

It is a linear feature extraction method that finds a new set of orthogonal axes (Principal Components) along which the variance of the data is maximised while retaining as much as useful information as possible. It is an unsupervised learning Algorithm.



Steps of PCA

* Step 1: Mean Centering of Data

$$X = \begin{matrix} & \begin{matrix} x_1 & x_2 \end{matrix} \\ \begin{matrix} 2 \\ 4 \\ 6 \end{matrix} & \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} \end{matrix}$$

Mean of Col 1 = $\mu_1 = 4$
" " " 2 = $\mu_2 = 3$

$$X_c = \begin{bmatrix} 2-4 & 1-3 \\ 4-4 & 3-3 \\ 6-4 & 5-3 \end{bmatrix} = \begin{bmatrix} -2 & -2 \\ 0 & 0 \\ 2 & 2 \end{bmatrix}$$

$$X_c = \begin{bmatrix} 2 & 1 \\ 4 & 3 \\ 6 & 5 \end{bmatrix} - \begin{bmatrix} 4 & 3 \\ 4 & 3 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 4 & 3 \\ 6 & 5 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 4 & 3 \end{bmatrix}$$

$$\boxed{X_c = X - 1\mu^T} \text{ where } \mu = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{bmatrix}_{p \times 1}$$

$$X = \underbrace{\begin{bmatrix} x_{11} & x_{12} & \dots & x_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{np} \end{bmatrix}}_{p \text{ features}} \left. \vphantom{\begin{bmatrix} x_{11} & x_{12} & \dots & x_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{np} \end{bmatrix}} \right\} n \text{ Rows}$$

$$1 = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}_{n \times 1}$$

* Step 2: Standardization (if Required)

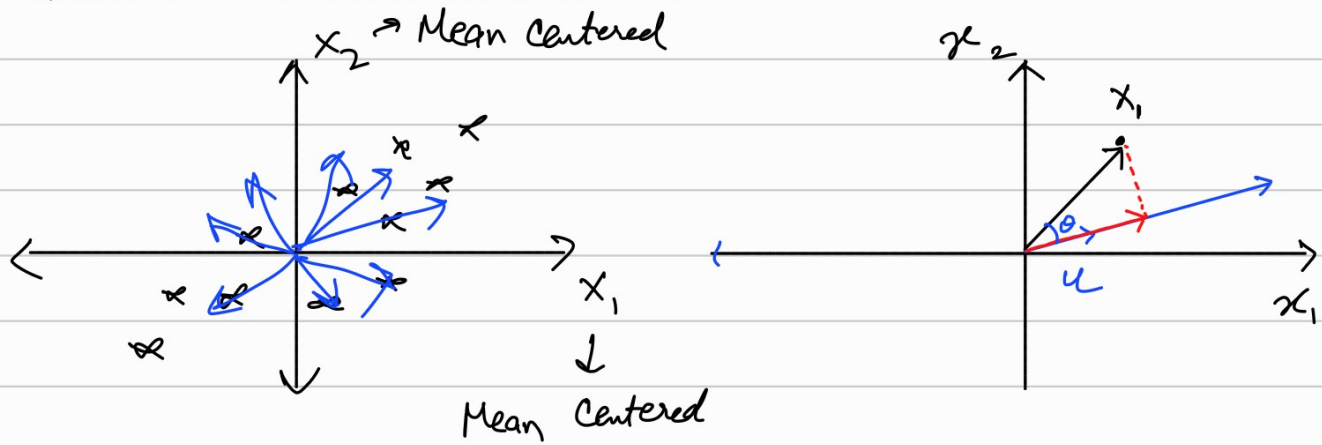
$$X \rightarrow Z = \frac{X - \mu}{\sigma}$$

$$\boxed{Z_{ij} = \frac{x_{ij} - \mu_j}{s_j}}$$

s_j = Sample std. deviation of j^{th} column.

$$\boxed{s_j^2 = \frac{1}{n-1} \sum_{i=1}^n (x_{ij} - \mu_j)^2}$$

Problem Formulation



u is the unit vector in one of the possible chosen vectors.

Length of Projection vector of x_1 on $u = |x_1| \cos \theta$

$$\cos \theta = \frac{\vec{u} \cdot \vec{x}_1}{|\vec{x}_1| \cdot |\vec{u}|} \quad \longrightarrow \quad = \vec{u} \cdot \vec{x}_1$$

$$|\vec{u}| = 1$$

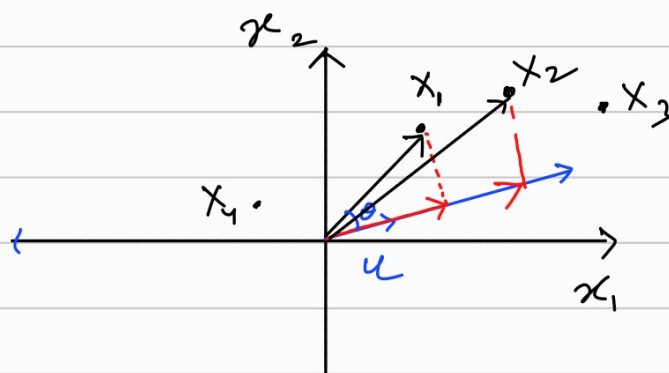
$$= u^T x_1$$

$$V_1 = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, V_2 = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$$

$$\vec{V}_1 \cdot \vec{V}_2 = x_1 z_1 + x_2 z_2$$

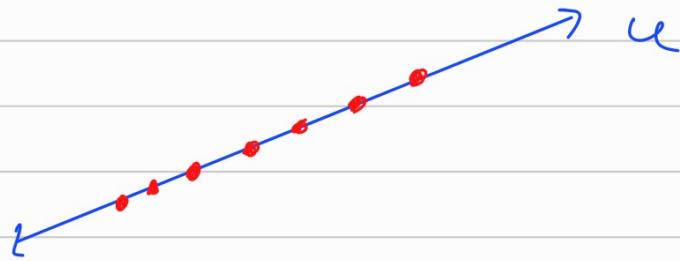
$$= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$$

$$= V_1^T V_2$$



Length of Projection of x_2 on $u = u^T x_2$

" " " " x_3 on $u = u^T x_3$



Variance of Projected lengths on u

$$= \frac{1}{n-1} \sum_{i=1}^n (u^T x_i - \overline{u^T x})^2$$

$$\begin{aligned} \overline{u^T x} &= \frac{1}{n} \sum_{i=1}^n u^T x_i = u^T \left(\frac{1}{n} \sum_{i=1}^n x_i \right) \\ &= u^T \bar{x} \end{aligned}$$

For centered Matrix $x_c \Rightarrow \bar{x} = 0$

$$\boxed{\text{Var} = \frac{1}{n-1} \sum_{i=1}^n (u^T x_i)^2}$$

I want to maximise this function with constraint that $\boxed{\|u\| = 1}$

We solve this using Lagrange Multiplier.

* We get an elegant solution:

Step 3: Find the covariance Matrix of features:

$$S = \begin{matrix} & \begin{matrix} f_1 & f_2 & \dots & f_p \end{matrix} \\ \begin{matrix} f_1 \\ f_2 \\ \vdots \\ f_p \end{matrix} & \begin{bmatrix} \text{Cov}(f_1, f_1) & \text{Cov}(f_1, f_2) & \dots & \text{Cov}(f_1, f_p) \\ \text{Cov}(f_2, f_1) & \text{Cov}(f_2, f_2) & \dots & \text{Cov}(f_2, f_p) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(f_p, f_1) & \dots & \dots & \text{Cov}(f_p, f_p) \end{bmatrix} \end{matrix}$$

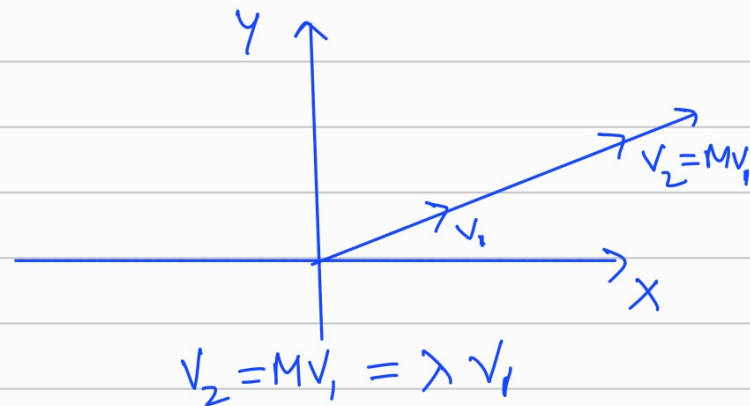
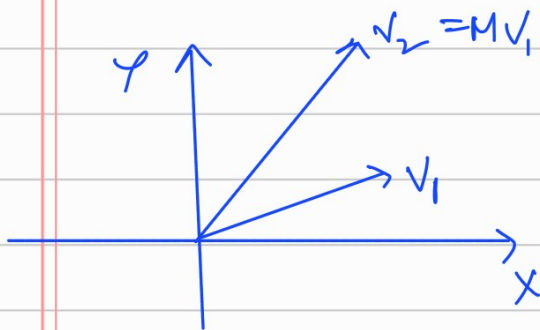
$$\text{Cov}(X, Y) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

$$\text{Cov}(f_j, f_k) = \frac{1}{n-1} \sum_{i=1}^n (x_{ij} - \mu_j)(x_{ik} - \mu_k)$$

Short form for covariance matrix, S :

$$S = \frac{1}{n-1} X_c^T X_c$$

Step 4: find the eigen values and eigen vectors of S :



v_1 is the eigen vector of M and λ is its eigen value.

$\Rightarrow$

$$S \vec{v}_1 = \lambda \vec{v}_1$$

$$S \vec{v}_1 - \lambda I \vec{v}_1 = 0 \Rightarrow (S - \lambda I) \vec{v}_1 = 0$$

when

$$|S - \lambda I| = 0$$

After solving this, you will get eigen vectors & eigen values of S .

$$\begin{array}{ccccccc} \lambda_1 & \lambda_2 & \dots & \lambda_p \\ \downarrow & \downarrow & & \downarrow \\ v_1 & v_2 & \dots & v_p \end{array}$$

Step 5: Sort these eigen values in descending order.

Lets say I have 3 eigen vector & values:

$$\begin{array}{l} \lambda_1 = 1.5 \\ \lambda_2 = 0.5 \\ \lambda_3 = 1.0 \end{array} \left. \vphantom{\begin{array}{l} \lambda_1 = 1.5 \\ \lambda_2 = 0.5 \\ \lambda_3 = 1.0 \end{array}} \right\} \begin{array}{ccc} \lambda_1 & & \lambda_3 & & \lambda_2 \\ \downarrow & & \downarrow & & \downarrow \\ v_1 & & v_3 & & v_2 \\ \downarrow & & \downarrow & & \downarrow \\ \text{PC1} & & \text{PC2} & & \text{PC3} \end{array}$$

More the eigen value, more variance will it carry along the direction of its eigen vector.

Find EVR for each eigen vector:

$$\begin{array}{l} \text{PC1} = \frac{\lambda_1}{\sum \lambda_i} \\ \text{(EVR)} \end{array}, \begin{array}{l} \text{PC2} = \frac{\lambda_2}{\sum \lambda_i} \\ \text{(EVR)} \end{array}$$

SECTION 1 From selecting features to extracting directions

1.1 Why dimensionality can become a learning problem

When a dataset has many features (columns), each observation can be thought of as a point in a high-dimensional space. More features mean more information for the model to process. However, not all features are equally useful. Some features may carry important patterns (signal), while others may add unnecessary complexity or random noise. The challenge is: how do we keep the useful information while reducing the unnecessary parts?

◆ TAKEAWAY

High dimensionality is not automatically bad; the difficulty arises when representation size grows faster than stable information.

1.2 Feature selection and feature extraction

Visual: Selection retains original axes; PCA constructs new axes

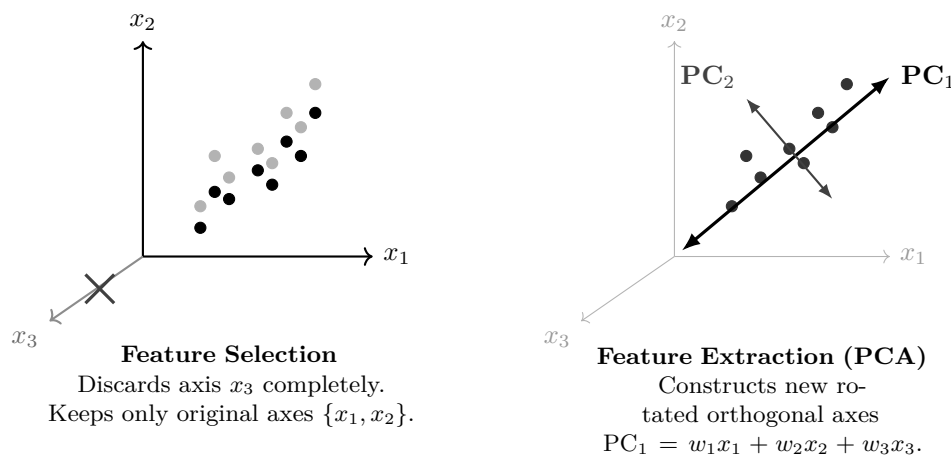


Fig. 1: Feature selection discards entire original features (axes), potentially losing signal. PCA constructs new rotated orthogonal axes that align with the true directions of maximum variance.

1.3 What is PCA?

What is PCA?

Imagine you have a dataset with many features, but most of the useful information is concentrated along just a few directions. PCA is a technique that finds these important directions in the data.

What problem does it solve?

When we have too many features, our models can become slow and may learn noise instead of real patterns. PCA reduces the number of features while keeping as much useful information as possible.

What is it actually doing to the data?

PCA rotates the coordinate axes of the data so that the new axes point along the directions where the data varies the most. It then keeps only the top few directions (called *principal components*) and discards the rest. Think of it as finding the best angle to look at the data so that you see the most spread.

★ KEY INSIGHT

Principal Component Analysis (PCA) is an unsupervised linear feature-extraction method that finds a new set of orthogonal axes (principal components) along which the variance of the data is maximized.

PART II: BUILDING BLOCKS FOR PCA

Before applying PCA, we need to understand a few key concepts: centering, projections, covariance, and eigenvectors.

SECTION 2 Building blocks for PCA

2.1 Data matrix and notation

Each row of X is one observation. Each column is one measured feature. PCA changes the coordinate system used to describe the same observations.

Symbol	Dimensions	Meaning
n	—	Number of observations
p	—	Number of original features
k	—	Number of retained principal components ($k \leq p$)
X	$n \times p$	Original data matrix; row i is observation $\mathbf{x}_i^\top$
$\boldsymbol{\mu}$	$p \times 1$	Vector of training-feature means
X_c	$n \times p$	Centered data matrix
S	$p \times p$	Sample covariance matrix of the centered features
$\mathbf{v}_j$	$p \times 1$	Unit eigenvector / direction of principal component j
λ_j	—	Eigenvalue associated with $\mathbf{v}_j$; variance captured
W_k	$p \times k$	Matrix whose columns are the first k eigenvectors
Z	$n \times k$	Transformed data: the principal-component scores

△ WARNING

Split first. Fit the means, standard deviations, and PCA directions on the **training set only**. Apply the same fitted transformation to validation and test data.

2.2 Step 1: Mean centering and standardization

The first step before applying PCA is to **mean center** the data. This means subtracting the average value of each feature from every observation:

$$X_c = X - \mathbf{1} \boldsymbol{\mu}^\top$$

Understanding every term and its dimensions:

- X is the **data matrix** of size $n \times p$ (n observations, p features).
- $\boldsymbol{\mu}$ is the **column mean vector** of size $p \times 1$, where $\mu_j = \frac{1}{n} \sum_{i=1}^n x_{ij}$ is the mean of feature j .
- $\boldsymbol{\mu}^\top$ is the **transpose** of $\boldsymbol{\mu}$, giving a $1 \times p$ **row vector**.
- $\mathbf{1}$ is a **column vector of ones** of size $n \times 1$: $\mathbf{1} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$.
- The product $\mathbf{1} \boldsymbol{\mu}^\top$ is an **outer product**: $(n \times 1)(1 \times p) = n \times p$. It replicates the mean row n times, producing a matrix the same size as X .

Numerical example (running dataset). Consider 3 houses ($n = 3$) with 2 features ($p = 2$). We will reuse this dataset throughout the worksheet:

House	Bedroom-area index f_1	Washroom-area index f_2
A	2	1
B	4	3
C	6	5

Feature means: $\mu_1 = \frac{2+4+6}{3} = 4$, $\mu_2 = \frac{1+3+5}{3} = 3$. So $\boldsymbol{\mu} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$ and $\boldsymbol{\mu}^\top = \begin{bmatrix} 4 & 3 \end{bmatrix}$.

$$\underbrace{\mathbf{1}}_{3 \times 1} \underbrace{\boldsymbol{\mu}^\top}_{1 \times 2} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 4 & 3 \end{bmatrix} = \underbrace{\begin{bmatrix} 4 & 3 \\ 4 & 3 \\ 4 & 3 \end{bmatrix}}_{3 \times 2}$$

Subtracting from X :

$$X_c = X - \mathbf{1} \boldsymbol{\mu}^\top = \begin{bmatrix} 2 & 1 \\ 4 & 3 \\ 6 & 5 \end{bmatrix} - \begin{bmatrix} 4 & 3 \\ 4 & 3 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} -2 & -2 \\ 0 & 0 \\ 2 & 2 \end{bmatrix}$$

After centering, every feature column has a mean of zero.

⚠ WARNING

Mean centering is compulsory. PCA finds directions of maximum variance. If the data is not centered, the mean offset distorts the variance calculation. Therefore, mean centering must **always** be performed before PCA, regardless of the dataset.

When is standardization also needed?

After mean centering, we may additionally need to **standardize** each feature by dividing by its standard deviation:

$$z_{ij} = \frac{x_{ij} - \mu_j}{s_j}$$

Here each symbol means:

- i = **observation index** (which row, i.e. which data point), ranging from 1 to n .
- j = **feature index** (which column, i.e. which variable), ranging from 1 to p .
- x_{ij} = the **original value** of feature j for observation i .
- μ_j = the **mean** of feature j across all n observations.
- s_j = the **sample standard deviation** of feature j : $s_j = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_{ij} - \mu_j)^2}$.

Continuing our running example: $s_1 = \sqrt{\frac{(-2)^2 + 0^2 + 2^2}{2}} = 2$, $s_2 = \sqrt{\frac{(-2)^2 + 0^2 + 2^2}{2}} = 2$.

$$z_{11} = \frac{2-4}{2} = -1, \quad z_{12} = \frac{1-3}{2} = -1, \quad z_{21} = \frac{4-4}{2} = 0, \quad z_{22} = \frac{3-3}{2} = 0$$

$$z_{31} = \frac{6-4}{2} = 1, \quad z_{32} = \frac{5-3}{2} = 1 \quad \implies \quad X_{\text{std}} = \begin{bmatrix} -1 & -1 \\ 0 & 0 \\ 1 & 1 \end{bmatrix}$$

- **Use centering only** when features share comparable units and their original variances carry meaningful information.
- **Use standardization** when features have different units or scales (e.g. rupees vs. years), so that no single feature dominates the variance purely due to its scale.

★ KEY INSIGHT

Summary: Mean centering is **always required** for PCA. Standardization is a **conditional choice** — apply it when features have different units or when you do not want any feature's original scale to dominate. In our running example, both features share the same units, so centering alone suffices.

2.3 When one direction captures most of the variance

Axis-aligned variance: Dropping x_2 preserves $> 98\%$ of total variance. Simple **Feature Selection** is sufficient!

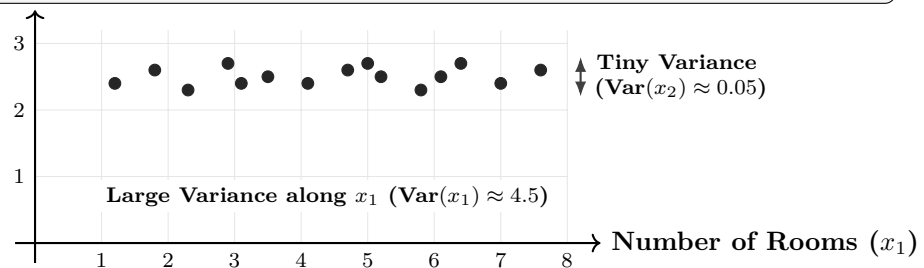


Fig. 2: When the data's variance aligns with an original feature axis, selecting that feature preserves nearly all information.

In this simple example, the data varies much more along the *Number of Rooms* direction than along the *Number of Grocery Shops* direction. If we had to choose only one feature, picking Number of Rooms would preserve most of the information.

But what happens when the data does not align neatly with any original axis?

2.4 Rotating the axes when data lies diagonally

Consider two housing measurements: bedroom area and washroom area. When both measurements increase together, the data cloud lies diagonally. Neither original axis follows the direction in which the data vary most.

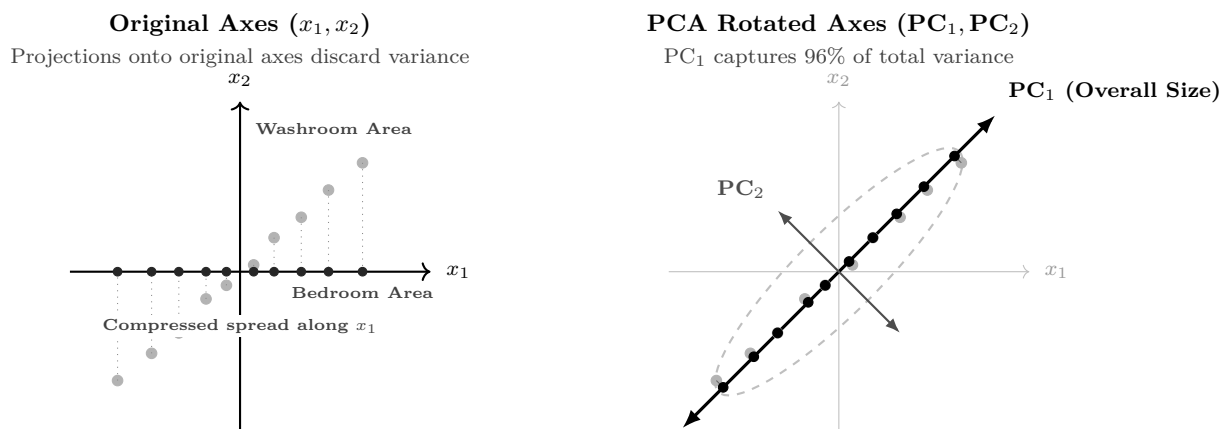


Fig. 3: Left: When features are correlated, projecting onto either original axis (x_1 or x_2) discards valuable variance. Right: PCA rotates the axes to align PC_1 directly along the diagonal spread, capturing nearly all information with a single component.

Unlike the previous example, neither original axis captures the main direction of spread. PCA finds the direction of maximum variance (PC_1) and rotates the coordinate system accordingly.

After this rotation, the entire dataset is transformed into the new coordinate system. If we keep only PC_1 , we reduce two features to one while retaining most of the information.

Important: The number of principal components is always less than or equal to the number of original features.

Geometric interpretation:

- PC_1 can represent an “overall house size” direction because bedroom and washroom area rise together.
- PC_2 is perpendicular and can represent the smaller contrast between how space is allocated to the two rooms.

2.5 Problem formulation: projection and variance

Let $\mathbf{u}$ be any direction in the original p -dimensional space. To use $\mathbf{u}$ as an axis, first make it a **unit vector** so that $\|\mathbf{u}\| = 1$. The scalar coordinate of centered observation $\mathbf{x}_i$ along $\mathbf{u}$ is:

$$z_i = \mathbf{u}^\top \mathbf{x}_i$$

Student exercise: Prove that the scalar coordinate of $\mathbf{x}_i$ along the unit direction $\mathbf{u}$ is given by the dot product $\mathbf{u}^\top \mathbf{x}_i$. Use the definition of projection onto a unit vector.

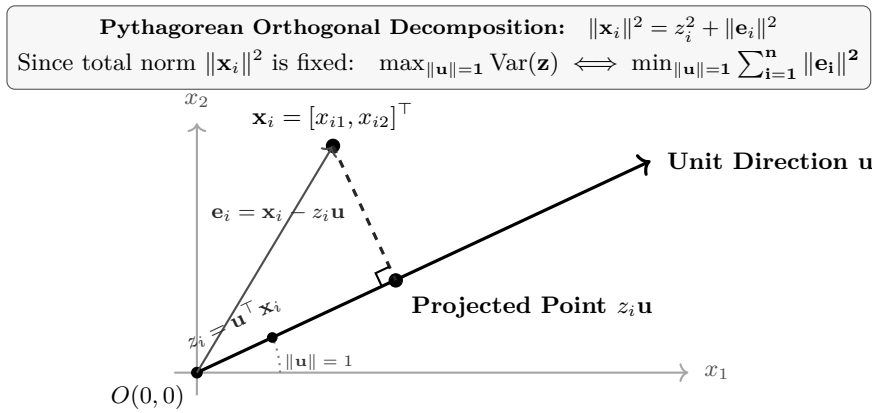


Fig. 4: The dot product $z_i = \mathbf{u}^\top \mathbf{x}_i$ assigns a 1D scalar coordinate to observation $\mathbf{x}_i$ along unit direction $\mathbf{u}$. Maximizing the variance of z_i is mathematically equivalent to minimizing the reconstruction error $\|\mathbf{e}_i\|^2$.

Suppose our centered dataset contains n observations: $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$.

When we project each observation onto the direction $\mathbf{u}$, we get n projection lengths:

$$\mathbf{u}^\top \mathbf{X}_1, \mathbf{u}^\top \mathbf{X}_2, \dots, \mathbf{u}^\top \mathbf{X}_n$$

The variance of these projection lengths measures how spread out the data is along $\mathbf{u}$:

$$\text{Variance along } \mathbf{u} = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{u}^\top \mathbf{X}_i - \overline{\mathbf{u}^\top \mathbf{X}})^2$$

where $\overline{\mathbf{u}^\top \mathbf{X}}$ is the mean of all projected values.

Simplification. Since projection is a *linear* operation, the mean of the projected values can be factored:

$$\overline{\mathbf{u}^\top \mathbf{X}} = \frac{1}{n} \sum_{i=1}^n \mathbf{u}^\top \mathbf{X}_i = \mathbf{u}^\top \left(\frac{1}{n} \sum_{i=1}^n \mathbf{X}_i \right) = \mathbf{u}^\top \bar{\mathbf{X}}$$

But we have already **centered** the data, so $\bar{\mathbf{X}} = \mathbf{0}$. Therefore:

$$\overline{\mathbf{u}^\top \mathbf{X}} = \mathbf{u}^\top \mathbf{0} = 0$$

The variance formula simplifies to:

$$\text{Variance along } \mathbf{u} = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{u}^\top \mathbf{X}_i)^2$$

Among the infinitely many possible unit vectors $\mathbf{u}$, our objective is to find the one that **maximizes** this variance. This is the core problem that PCA solves.

Solving this optimization problem mathematically is beyond the scope of this course. However, the solution turns out to be remarkably elegant: the optimal directions are the **eigenvectors** of the **covariance matrix**, and the corresponding **eigenvalues** tell us how much variance each direction captures.

Before we can apply this result, let us briefly review these key concepts.

2.6 Refresher: covariance matrix, eigenvalues, and eigenvectors

Before discussing the covariance *matrix*, let us first recall what **covariance** itself measures.

Covariance: how two centered features move together

For two features x and y measured on n observations:

$$\text{Cov}(x, y) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

- **Positive covariance:** Above-average x values tend to appear with above-average y values $\Rightarrow$ both features rise together.
- **Negative covariance:** Above-average x values tend to appear with below-average y values $\Rightarrow$ one rises while the other falls.
- **Near-zero covariance:** Little *linear* co-movement. A nonlinear relationship may still exist.

Using our running example (centered data X_c), the covariance between f_1 and f_2 is:

$$\text{Cov}(f_1, f_2) = \frac{(-2)(-2) + (0)(0) + (2)(2)}{3-1} = \frac{4+0+4}{2} = 4$$

We will explore covariance in greater depth in Section 3. For now, this definition is sufficient to understand the covariance matrix.

The Covariance Matrix

The **covariance matrix** S is a $p \times p$ matrix that summarizes how every pair of features varies together.

Anatomy of a three-feature covariance matrix:

	f_1	f_2	f_3
f_1	$\text{Var}(f_1)$	$\text{Cov}(f_1, f_2)$	$\text{Cov}(f_1, f_3)$
f_2	$\text{Cov}(f_2, f_1)$	$\text{Var}(f_2)$	$\text{Cov}(f_2, f_3)$
f_3	$\text{Cov}(f_3, f_1)$	$\text{Cov}(f_3, f_2)$	$\text{Var}(f_3)$

- **Diagonal entries** S_{jj} : the _____ of each feature.
- **Off-diagonal entries** S_{jk} : the _____ between features j and k .
- **Symmetry:** $S_{jk} = S_{kj}$, so the matrix mirrors across the diagonal.

Eigenvalues and Eigenvectors

For a square matrix A , an **eigenvector** $\mathbf{v}$ is a non-zero vector whose direction is preserved under the transformation — it only gets scaled by a factor λ (the **eigenvalue**):

$$A\mathbf{v} = \lambda\mathbf{v}$$

The eigenvector $\mathbf{v}$ points in a direction that the matrix A does not rotate. The eigenvalue λ tells us how much the matrix stretches or shrinks that direction.

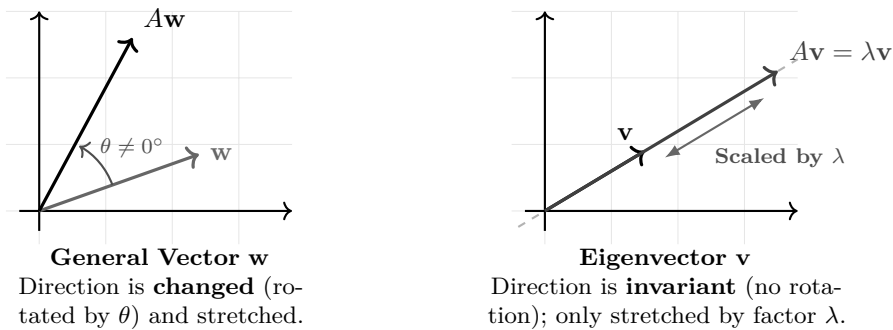


Fig. 5: Unlike a general vector which rotates under a matrix transformation A , an eigenvector v stays along its original line and is merely scaled by its eigenvalue λ .

Connection to PCA: In PCA, the matrix A is the covariance matrix S . The eigenvectors of S give us the principal directions, and the eigenvalues tell us the variance captured along each direction.

PART III: UNDERSTANDING COVARIANCE

Variance describes how one feature spreads around its mean. Covariance extends the idea to two features.

SECTION 3 Understanding covariance

3.1 Covariance: how two centered features move together

$$\text{Cov}(x, y) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

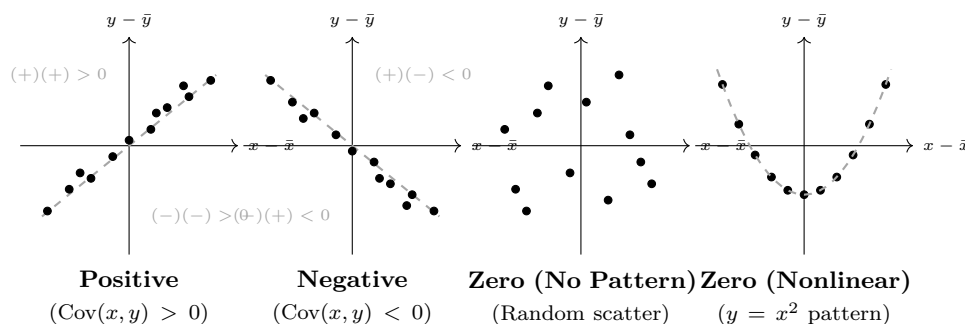


Fig. 6: Panel 1–2: Positive and negative linear covariance. Panel 3: Random scatter with zero covariance and no relationship. Panel 4: A symmetric parabola also has $\text{Cov}(x, y) = 0$ despite a perfect deterministic nonlinear relationship — zero covariance does not imply independence.

- **Positive:** Above-average x values tend to appear with above-average y values.
- **Negative:** Above-average x values tend to appear with below-average y values.
- **Near-zero:** Little linear co-movement. A nonlinear relationship can still exist.

Covariance depends on units and is not bounded between -1 and $+1$. Correlation divides covariance by the two standard deviations.

Classroom question: Does zero covariance prove that two variables are independent?

PART IV: FINDING THE PRINCIPAL COMPONENTS

Now that we understand the building blocks, we can see how PCA finds the principal components.

SECTION 4 Finding the principal components

4.1 Eigenvalues and eigenvectors of the covariance matrix

The principal components are obtained by finding the eigenvalues and eigenvectors of the covariance matrix S . These satisfy:

$$S\mathbf{v} = \lambda\mathbf{v}$$

or equivalently, the eigenvalues can be found by solving:

$$|S - \lambda I| = 0$$

where S is the covariance matrix, λ is an eigenvalue, $\mathbf{v}$ is the corresponding eigenvector, and I is the identity matrix.

4.2 Ordering the principal components

Suppose for a dataset with 3 features, we compute the covariance matrix and obtain:

- Eigenvalues: $\lambda_1, \lambda_2, \lambda_3$
- Corresponding eigenvectors: $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$

We sort the eigenvalues in **descending order**. For example, if $\lambda_1 > \lambda_3 > \lambda_2$, then:

Principal Component	Direction	Variance Captured
PC ₁	$\mathbf{v}_1$	λ_1 (largest)
PC ₂	$\mathbf{v}_3$	λ_3 (second largest)
PC ₃	$\mathbf{v}_2$	λ_2 (smallest)

The eigenvector with the largest eigenvalue becomes PC₁ because it captures the most variance.

4.3 Selecting and projecting

The next step is to select the top k principal components that capture most of the data's variance.

To transform each observation into the new coordinate system, we project it onto the selected principal components:

$$z_i = \mathbf{u}^\top \mathbf{X}_i$$

where $\mathbf{u}$ is the unit eigenvector (principal direction) and $\mathbf{X}_i$ is the centered observation.

This projection is performed for **every observation** in the dataset and for **each retained principal component**.

★ KEY INSIGHT

Each eigenvalue of the covariance matrix tells us exactly how much variance is captured along its corresponding principal direction. Larger eigenvalue = more information retained.

4.4 A small numerical example

Three houses described by two already comparable measurements:

House	Bedroom-area index f_1	Washroom-area index f_2
A	2	1
B	4	3
C	6	5

Step 1 — Center the columns. The mean vector is $\boldsymbol{\mu} = [4, 3]^\top$:

$$X_c = \begin{bmatrix} -2 & -2 \\ 0 & 0 \\ 2 & 2 \end{bmatrix}$$

Step 2 — Compute the sample covariance matrix.

Why does $S = \frac{1}{n-1} X_c^\top X_c$ give the covariance matrix?

Recall that the sample covariance between features j and k is:

$$S_{jk} = \frac{1}{n-1} \sum_{i=1}^n (x_{ij} - \mu_j)(x_{ik} - \mu_k)$$

Now consider the matrix product $X_c^\top X_c$. Since X_c is $n \times p$, its transpose $X_c^\top$ is $p \times n$, and the product $X_c^\top X_c$ is $p \times p$. The (j, k) entry of this product is:

$$(X_c^\top X_c)_{jk} = \sum_{i=1}^n (X_c)_{ij} (X_c)_{ik} = \sum_{i=1}^n (x_{ij} - \mu_j)(x_{ik} - \mu_k)$$

This is exactly the numerator of S_{jk} ! Dividing by $(n-1)$ gives:

$$\frac{1}{n-1} (X_c^\top X_c)_{jk} = S_{jk}$$

Therefore, $S = \frac{1}{n-1} X_c^\top X_c$ produces the entire covariance matrix in one matrix operation.

Applying to our running example ($n = 3$, so $n - 1 = 2$):

$$S = \frac{1}{2} X_c^\top X_c = \frac{1}{2} \begin{bmatrix} -2 & 0 & 2 \\ -2 & 0 & 2 \end{bmatrix} \begin{bmatrix} -2 & -2 \\ 0 & 0 \\ 2 & 2 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix} = \underline{\hspace{2cm}}$$

Step 3 — Find eigenvalues and eigenvectors. Solve $|S - \lambda I| = 0$:

Component	Unit eigenvector	Eigenvalue	Meaning
PC ₁	$\mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$	$\lambda_1 = \underline{\hspace{2cm}}$	Both features increase together.
PC ₂	$\mathbf{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$	$\lambda_2 = \underline{\hspace{2cm}}$	No remaining contrast variation.

Step 4 — Project onto PC₁:

$$\mathbf{z}_1 = X_c \mathbf{v}_1 = \underline{\hspace{2cm}}$$

Step 5 — Interpret. PC₁ captures $\frac{8}{8+0} = \underline{\hspace{2cm}}$ of the total variance. The two features can be represented by one component with no reconstruction loss because all points lie on a .

Classroom question: What real-world phrase could describe PC₁ in this example?

PRACTICE P1 · FOUNDATIONS AND GEOMETRY

- (a) **Selection or Extraction?** For each scenario, decide whether the problem calls for **feature selection** or **PCA / feature extraction**.

Scenario	Decision	Reason
CGPA_copy exactly duplicates CGPA.	_____	_____
Bedroom area and washroom area are strongly correlated and both reflect overall house size.	_____	_____
A rare binary medical flag has low variance but is strongly associated with the target.	_____	_____
A 500-dim image embedding must be plotted for human inspection.	_____	_____

- (b) **Read covariance from geometry.** For each description, state the expected sign of covariance.

Description	Expected sign	Explanation
Study hours $\uparrow \Rightarrow$ practice-test score $\uparrow$	_____	_____
Product price $\uparrow \Rightarrow$ units sold $\downarrow$	_____	_____
Points form a symmetric U-shape around $x = 0$.	_____	_____

PRACTICE P2 · MATHEMATICAL CONCEPTS

(Homework — to be discussed in lab)

- (a) **Compute one projection.** Let $\mathbf{x} = [3, 1]^\top$ and $\mathbf{u} = \frac{1}{\sqrt{2}}[1, 1]^\top$.

- Verify that $\mathbf{u}$ is a unit vector: $\|\mathbf{u}\| =$ _____
- Compute the score: $z = \mathbf{u}^\top \mathbf{x} =$ _____
- Compute the projected point: $z \mathbf{u} =$ _____
- Explain why the projected point does not have to equal $\mathbf{x}$:

- (b) **Complete a miniature PCA.** The centered observations are $\mathbf{x}_1 = [-1, -1]^\top$, $\mathbf{x}_2 = [0, 0]^\top$, $\mathbf{x}_3 = [1, 1]^\top$.

- (i) Form X_c and compute $S = \frac{1}{n-1} X_c^\top X_c$:

$$X_c = \text{_____}, \quad S = \text{_____}$$

- Show that $\mathbf{v}_1 = \frac{1}{\sqrt{2}}[1, 1]^\top$ has eigenvalue _____.
- Show that $\mathbf{v}_2 = \frac{1}{\sqrt{2}}[1, -1]^\top$ has eigenvalue _____.
- Explained variance ratio of PC₁: _____. Lossless? _____

PART V: THE COMPLETE PCA PROCEDURE AND PRACTICAL DECISIONS

PCA has been explained. Now organize the procedure end-to-end and address choosing k .

SECTION 5 The complete PCA procedure and practical decisions

5.1 Step-by-step PCA workflow

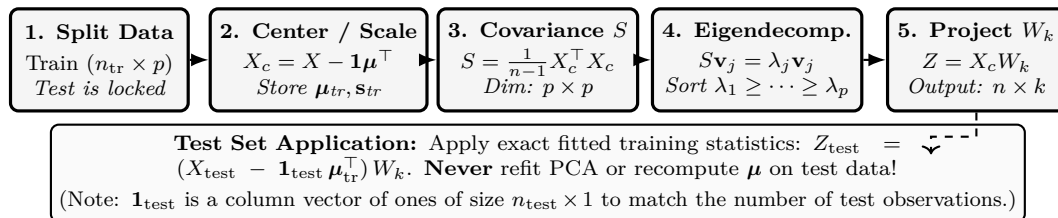


Fig. 7: The complete end-to-end PCA machine learning pipeline. All statistics (μ, s) and projection directions (W_k) are learned exclusively on the training set and applied unchanged to test/production data.

Note: Stage 2 uses the formula $X_c = X - \mathbf{1}\mu^\top$. Please refer back to **Section 2.2** for a detailed step-by-step numerical example explaining the dimensions of this matrix subtraction.

1. Create the data split before learning any preprocessing. The test set remains untouched.
2. Fit centering (and, when appropriate, standardization) on the **training features**. Store μ and s .
3. Construct the covariance matrix S from the centered training data.
4. Find eigenvalues and unit eigenvectors of S .
5. Sort eigenpairs by descending eigenvalue.
6. Choose k using explained variance, visualization needs, and downstream validation.
7. Form $W_k = [\mathbf{v}_1 \cdots \mathbf{v}_k]$. Its dimensions are _____.
8. Transform: $Z_{train} = X_{c,train} W_k$ and $Z_{test} = (X_{test} - \mathbf{1}_{test} \mu_{tr}^\top) W_k$, using **stored training** quantities. (Here $\mathbf{1}_{test}$ is of size $n_{test} \times 1$.)

5.2 How do we determine the optimum number of principal components?

Recall that each eigenvalue λ_j of the covariance matrix represents the **amount of variance** captured by the corresponding principal component. A large eigenvalue means that direction captures a lot of the data's spread; a small eigenvalue means that direction captures very little.

The **total variance** in the dataset is simply the sum of all eigenvalues:

$$\text{total variance} = \sum_{j=1}^p \lambda_j = \text{trace}(S)$$

The **Explained Variance Ratio (EVR)** tells us what fraction of the total variance each principal component captures:

$$\text{EVR of PC}_j = \frac{\lambda_j}{\sum_{\ell=1}^p \lambda_\ell}, \quad \text{cumulative EVR through } k = \frac{\sum_{j=1}^k \lambda_j}{\sum_{\ell=1}^p \lambda_\ell}$$

Visual: Scree plot with cumulative explained variance

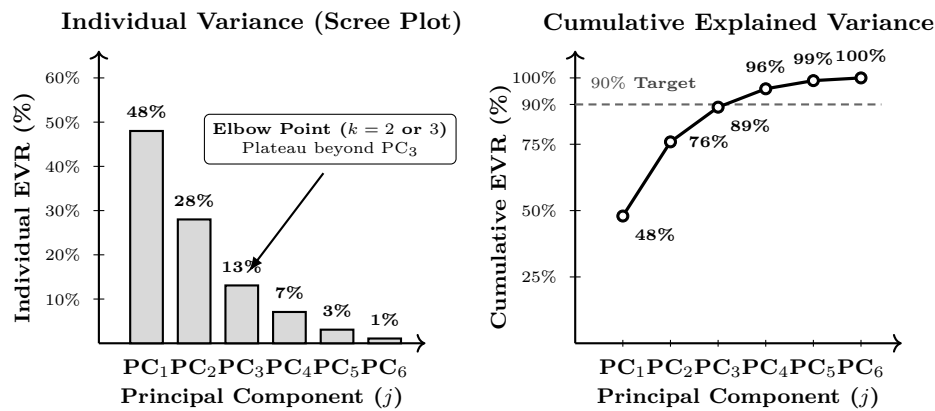


Fig. 8: A scree plot displays individual eigenvalues (bars) and cumulative explained variance (line). The “elbow” indicates where additional components yield diminishing returns.

Rules for choosing k :

- **Cumulative threshold:** Choose the smallest k reaching 90% or 95%, then validate.
- **Elbow / scree rule:** Look for the point after which eigenvalues flatten.
- **Visualization:** Choose $k = 2$ or 3 for human inspection.
- **Downstream prediction:** Treat k as a hyperparameter inside cross-validation.

Classroom question: If the first two components capture 76% of total variance, is $k = 2$ automatically the correct choice?

5.3 When PCA does not perform well

PCA works by finding directions of maximum variance. However, there are situations where this approach fails:

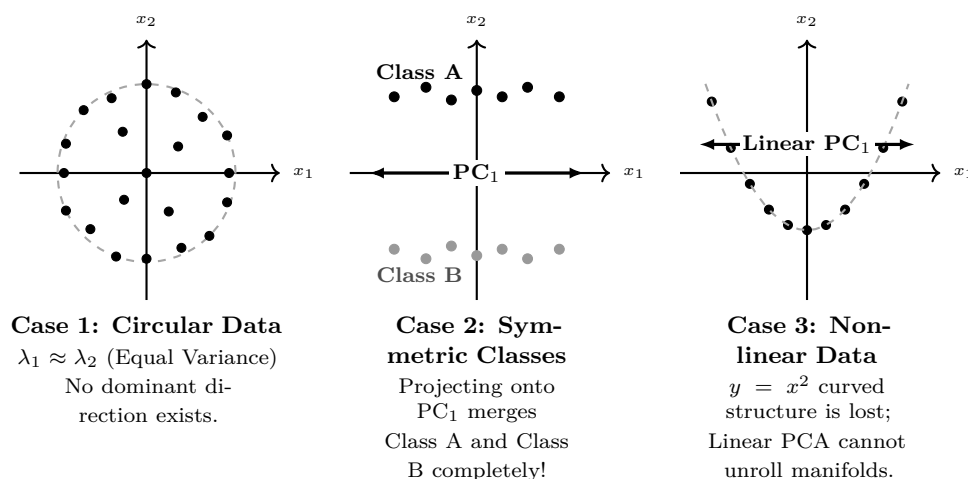


Fig. 9: Key failure cases of linear PCA. Left: Isotropic spread yields arbitrary components. Middle: Unsupervised variance maximization can destroy class separability. Right: Linear projections cannot capture curved nonlinear manifolds.

5.4 What PCA changes — and what it can fail to preserve

PCA can help with:

- Visualization and compression
- Faster downstream learning

- Removal of linear redundancy
- Noise reduction when low-variance directions are mostly noise

PCA does not guarantee:

- Better prediction
- Causal meaning or class separation
- Preservation of a low-variance feature that is highly predictive of y

△ WARNING

Do not fit scaling or PCA before cross-validation and then cross-validate the downstream model. Fit PCA inside each fold on the training portion only.

5.5 Comparison with feature selection

Criterion	Feature Selection	Feature Extraction (PCA)
What is reduced?	Number of original columns retained	Number of new component coordinates
Are original means preserved?	Yes	Usually not directly
Treatment of correlation	Often keeps one representative, removes another	Combines shared variation into one direction
Uses the target y ?	Depends on the method	No — standard PCA is unsupervised
Primary mathematical idea	Keep or discard columns	Rotate the coordinate system, then project

◆ TAKEAWAY

Feature selection asks which original axes deserve to remain. PCA asks whether the apparent complexity of the feature space can be summarized by a smaller set of rotated axes that still capture the relevant structure.

PRACTICE P3 · DIAGNOSIS AND APPLICATION*(Homework — to be discussed in lab)*

(a) **Choose k from eigenvalues.** Suppose the eigenvalues are $[6, 2, 1, 0.5]$.

- (i) Compute total variance: _____
 (ii) Complete the table:

k	$\sum_{j=1}^k \lambda_j$	Cumulative ratio	$\geq 90\%$?
1	_____	_____	_____
2	_____	_____	_____
3	_____	_____	_____
4	_____	_____	_____

- (iii) Smallest k that retains at least 90%: _____
 (iv) Explain why that k is still only a candidate for a predictive pipeline:

(b) **Diagnose the pipeline.** A student standardizes the full dataset, fits PCA once using all observations, keeps the first five components, and then reports five-fold cross-validation accuracy.

- (i) Is the reported score trustworthy? _____
 (ii) Which steps have seen the validation folds?

(iii) Describe the correct fold-by-fold procedure:

PRACTICE P4 · COMPREHENSIVE CHALLENGE*(Homework — to be discussed in lab)*

(a) **Fill in the Key Terms table:**

Term	Definition
Centering	_____
Covariance matrix	_____
Eigenvector	_____
Eigenvalue	_____
Principal component	_____
Explained variance	_____
Scree plot	_____

(b) **True or False:**

Statement	T/F	Justification
PCA uses the target variable y to find directions.	[]	_____
Principal components are always uncorrelated.	[]	_____
Standardization is always necessary before PCA.	[]	_____
The eigenvalue of a PC equals the variance of the scores along that direction.	[]	_____
PCA can unfold a curved manifold.	[]	_____

(c) **Synthesis questions.** Answer any three:

- (i) Why is centering essential before computing the covariance matrix?
- (ii) What does an eigenvalue mean geometrically in PCA?
- (iii) Why are principal-component scores uncorrelated?
- (iv) When can standardization change the first principal component dramatically?
- (v) Why can PCA improve a model in one dataset and harm it in another?
- (vi) How does PCA connect back to the bias–variance tradeoff and feature selection?
- (vii) Give an example of a dataset where PCA would fail to capture the true structure.

BY THE END OF THIS SHEET, YOU CAN

- ☐ Distinguish feature selection from feature extraction and state when PCA is appropriate.
- ☐ Write the centering formula and explain when standardization is needed.
- ☐ Explain what the covariance matrix records and identify its diagonal and off-diagonal entries.
- ☐ Explain how eigenvalues and eigenvectors of the covariance matrix yield the principal components.
- ☐ State why the principal directions are eigenvectors of the covariance matrix and why each eigenvalue represents the variance captured.
- ☐ Project data onto a unit direction and interpret the score geometrically.
- ☐ Complete a miniature PCA calculation by hand.
- ☐ Use explained variance ratios and a scree plot to choose k .
- ☐ Identify situations where PCA does not perform well.
- ☐ Explain why PCA must occur inside cross-validation folds.
- ☐ Compare PCA with feature selection across purpose, supervision, and validation discipline.

Worksheet 11 · AML · Lecture 11: Dimensionality Reduction and PCA.